Risk Analysis Report

1. Executive Summary

TITLE:

Portfolio Risk Distribution Overview

WHAT I DID

Analyzed the distribution of vendors across different risk levels to understand overall portfolio exposure.

```sql
-- 1. EXECUTIVE SUMMARY (High-Level Profile)
-- Summary of total vendors, average risk scores, and overall status.
SELECT
    Overall_risk_complaince AS Risk_Level,
    COUNT(*) AS Vendor_Count,
    ROUND((COUNT(*) / (SELECT COUNT(*) FROM vendor_audit)) * 100, 2) AS Percentage_of_Portfolio
FROM vendor_audit
GROUP BY Overall_risk_complaince
ORDER BY Field(Overall_risk_complaince, 'High', 'Medium', 'Low');
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Risk_Level	Vendor_Count	Percentage_of_Portfolio
Critical	260	26.00
High	221	22.10
Medium	257	25.70
Low	262	26.20

INSIGHT

This provides a high-level view of vendor risk distribution, helping stakeholders understand how risk is spread across the entire vendor portfolio.

2. Weighted Risk Scoring

TITLE:

Weighted Vendor Risk Scoring Model

WHAT I DID

Calculated a weighted risk index using cyber (50%), financial (30%), and compliance (20%) scores to rank vendor risk exposure.

Risk Analysis Report

```sql
-- 2. WEIGHTED RISK SCORINGtprm_vendors
-- Formula: Cyber (50%), Financial (30%), Compliance (20%)
SELECT
    vendor_name,
    industry,
    ROUND((cyber_score * 0.5) + (Financial_score * 0.3) + (Compliance_score * 0.2), 2) AS Weighted_Risk_Index,
    CASE
        WHEN (cyber_score * 0.5) + (Financial_score * 0.3) + (Compliance_score * 0.2) > 75 THEN 'Elite Partner'
        WHEN (cyber_score * 0.5) + (Financial_score * 0.3) + (Compliance_score * 0.2) BETWEEN 50 AND 75 THEN 'Standard'
        ELSE 'High-Exposure / Remediation Required'
    END AS Vendor_Tier
FROM v_final_risk_report
ORDER BY Weighted_Risk_Index DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Vendor_name	industry	Weighted_Risk_Index	Vendor_Tier
Mydo	Logistics	94.7	Elite Partner
Mydo	Logistics	94.0	Elite Partner
Mydo	Logistics	91.0	Elite Partner
Mydo	Logistics	90.2	Elite Partner
Mydo	Logistics	89.8	Elite Partner
Mydo	Logistics	89.6	Elite Partner
Mydo	Logistics	89.6	Elite Partner
Mydo	Logistics	89.4	Elite Partner
Mydo	Logistics	89.0	Elite Partner
Mydo	Logistics	88.8	Elite Partner
Mydo	Logistics	88.3	Elite Partner
Mydo	Logistics	88.0	Elite Partner
Mydo	Logistics	87.6	Elite Partner
Mydo	Logistics	87.5	Elite Partner
Mydo	Logistics	86.6	Elite Partner
Mydo	Logistics	86.3	Elite Partner

INSIGHT

This model enables prioritization of vendors based on overall risk exposure, allowing decision-makers to focus on high-impact risk areas.

3. Industry Criticality Analysis

TITLE:

Industry-Level Risk Assessment

WHAT I DID

Analyzed average risk scores across industries to determine which sectors pose the highest risk exposure.

Risk Analysis Report

```sql
-- 3. INDUSTRY CRITICALITY ANALYSIS
-- Identifying which sectors carry the most weight.
SELECT
    v.Category,
    COUNT(v.vendor_id) AS Total_Vendors,
    AVG(a.cyber_score) AS Avg_Cyber_Score,
    AVG(a.Financial_score) AS Avg_Financial_Score,
    CASE
        WHEN AVG(a.cyber_score) < 60 THEN 'IMMEDIATE ATTENTION'
        WHEN AVG(a.cyber_score) BETWEEN 60 AND 75 THEN 'MONITOR'
        ELSE 'STABLE'
    END AS Strategic_Priority
FROM tprm_vendors v
JOIN vendor_audit a ON v.vendor_id = a.vendor_id
GROUP BY v.Category
ORDER BY Avg_Cyber_Score ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Category	Total_Vendors	Avg_Cyber_Score	Avg_Financial_Score	Strategic_Priority
Logistics	1000	50.8690	51.5780	IMMEDIATE ATTENTION

INSIGHT

Certain industries show consistently lower cyber scores, indicating higher strategic risk and the need for tighter vendor controls.

4. Red Flag Report

TITLE:

High-Risk Vendor Identification (Red Flag Report)

WHAT I DID

Identified vendors in critical categories with low cyber scores (< 50) and high risk classification

Risk Analysis Report

```sql
--  THE "RED FLAG" REPORT (Risk Concentration)
-- Vendors that could cause a bankruptcy or a data breach tomorrow.
SELECT
    v.vendor_name,
    v.category,
    a.cyber_score,
    a.Compliance_score,
    a.Overall_risk_complaince
FROM tprm_vendors v
JOIN vendor_audit a ON v.vendor_id = a.vendor_id
WHERE v.category = 'Critical'
  AND (a.cyber_score < 50 OR a.Overall_risk_complaince = 'High')
ORDER BY a.cyber_score ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

vendor_name	category	cyber_score	Compliance_score	Overall_risk_complaince

INSIGHT

These vendors represent immediate operational and security threats and should be prioritized for remediation or contract review, fortunately there is zero threat

5. Remediation Tracker

TITLE

Vendor Remediation Action Plan

WHAT I DID

Generated actionable remediation steps based on vendor risk scores to guide corrective actions.

Risk Analysis Report

```sql
-- . THE REMEDIATION TRACKER
-- Queries that identify which vendors need a "Fix-it" plan.
SELECT
    v.vendor_name,
    a.Overall_risk_complaince,
    'Urgent' AS action_priority,
    CASE
        WHEN a.cyber_score < 40 THEN 'Initiate Technical Security Audit'
        WHEN a.Financial_score < 40 THEN 'Request 3-Year Audited Financials'
        ELSE 'Quarterly Performance Review'
    END AS consultant_next_steps
FROM tprm_vendors v
JOIN vendor_audit a ON v.vendor_id = a.vendor_id
WHERE a.Overall_risk_complaince = 'High';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

vendor_name	Overall_risk_complaince	action_priority	consultant_next_steps
Mydo	High	Urgent	Quarterly Performance Review
Mydo	High	Urgent	Initiate Technical Security Audit
Mydo	High	Urgent	Initiate Technical Security Audit
Mydo	High	Urgent	Quarterly Performance Review
Mydo	High	Urgent	Initiate Technical Security Audit
Mydo	High	Urgent	Quarterly Performance Review
Mydo	High	Urgent	Quarterly Performance Review
Mydo	High	Urgent	Quarterly Performance Review

INSIGHT

This transforms raw risk data into actionable business decisions, supporting governance and risk mitigation strategies.